Ghazi University D.G. Khan

Assignment

Submitted from:-

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Section:- 'E'

Subject:- Programming Fundamental

Semester:- 1st semester

Department:- CS

Morning.

Submitted To:-

Sir Asad Majeed

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Lab No 5

Task 1: Table of any number

```
#include<iostream>
using namespace std;
int main(){
    int num;
    cout<<"Enter any number:";
    cin>>num;
    for(int i=1;i<=10;i++){
        cout<<num<<" * "<<i<<" = "<<num*i
        <<endl;
    }
    return 0;
}
```

Output:-

Enter any number.

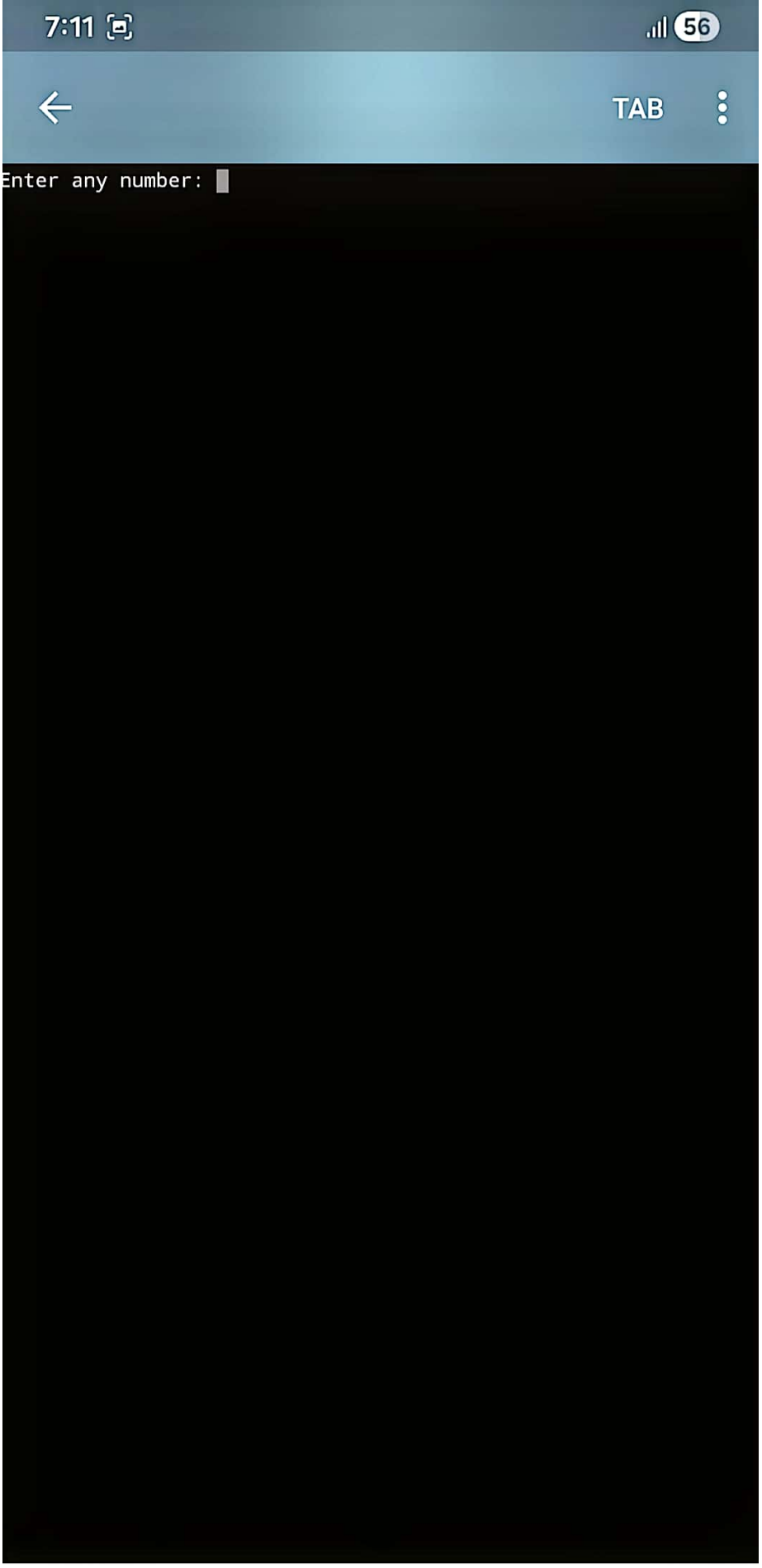
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new*

```
1  #include <iostream>
2  using namespace std;
3  int main() {
4      int num;
5      cout << "Enter any number: ";
6      cin >> num;
7      for(int i = 1; i <= 10; i++) {
8          cout << num << " * " << i << " = " << num * i
   << endl;
9      }
10     return 0;
11 }
12
```

Tab | { | } | : | ; | "

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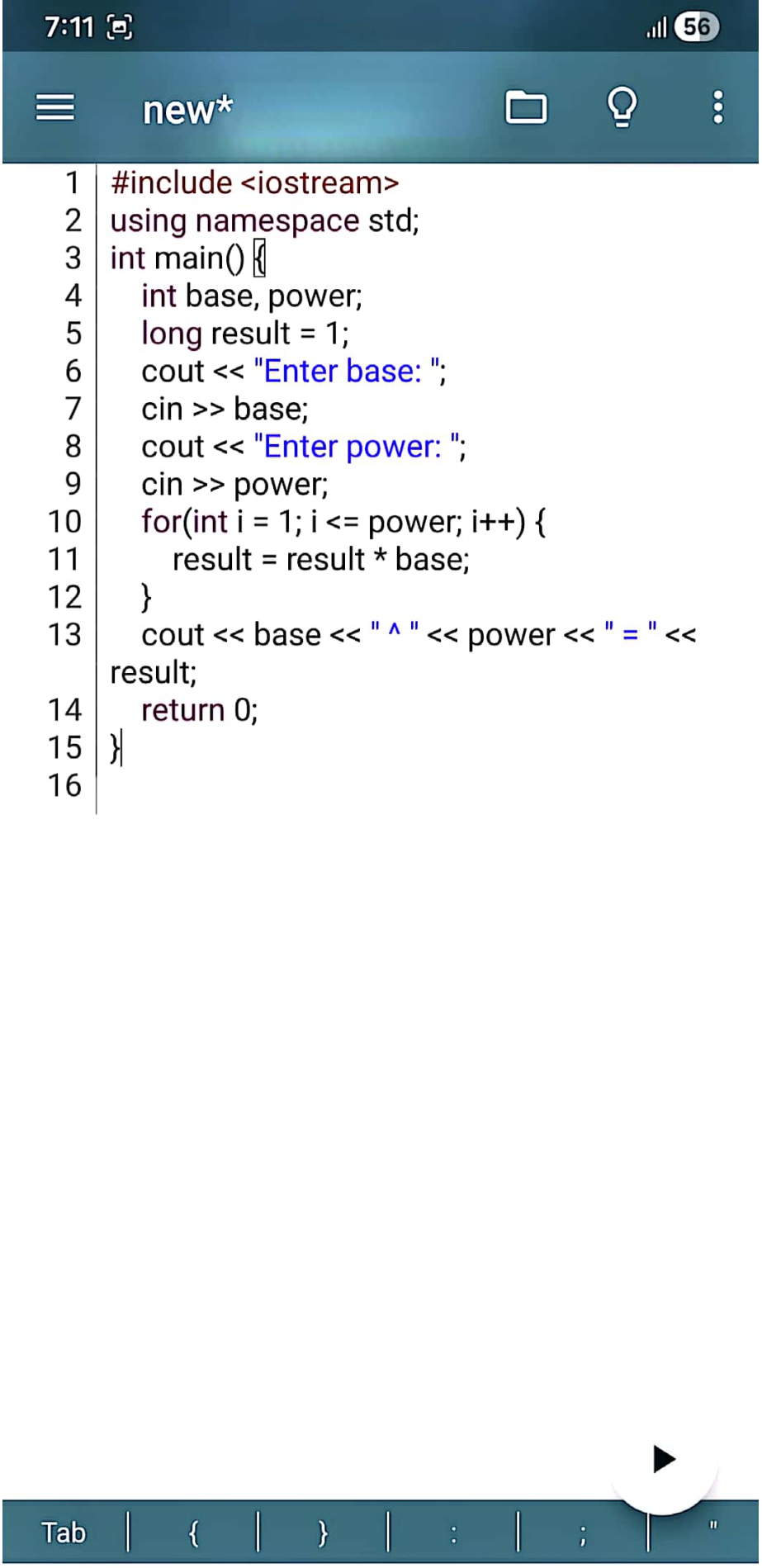
TASK 2: Power of any number using for loop

```
#include <iostream>
using namespace std;
int main( ){
    int base, Power;
    long result = 1;
    cout << "Enter base:=";
    cin >> base;
    cout << "Enter Power:=";
    cin >> power;
    for(int i=1; i<=power; i++){
        result=result* base;
    }
    cout<<base << "^" <<power << "=" <<result;
    return 0;
}
```

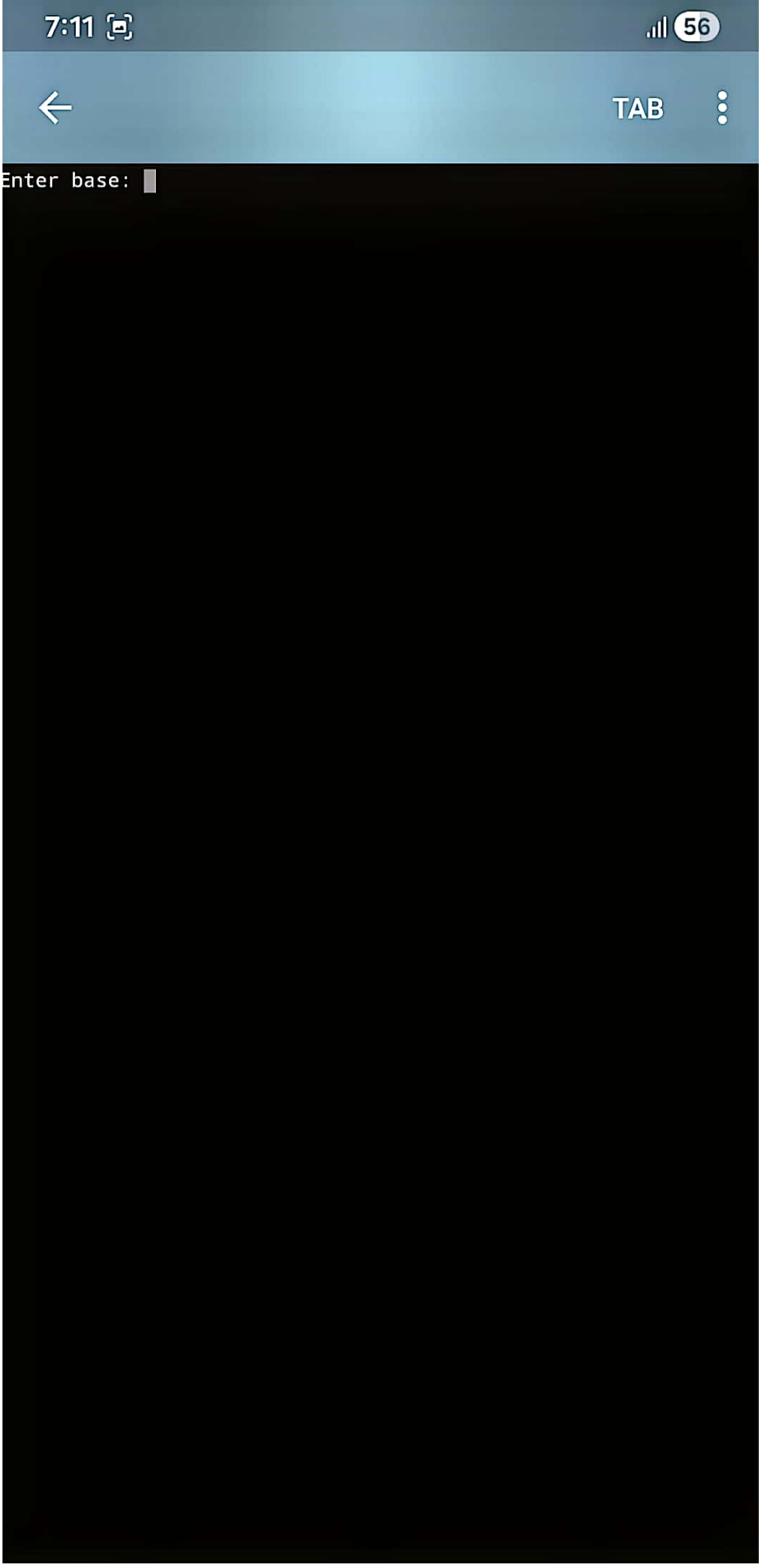
output:-

Enter base

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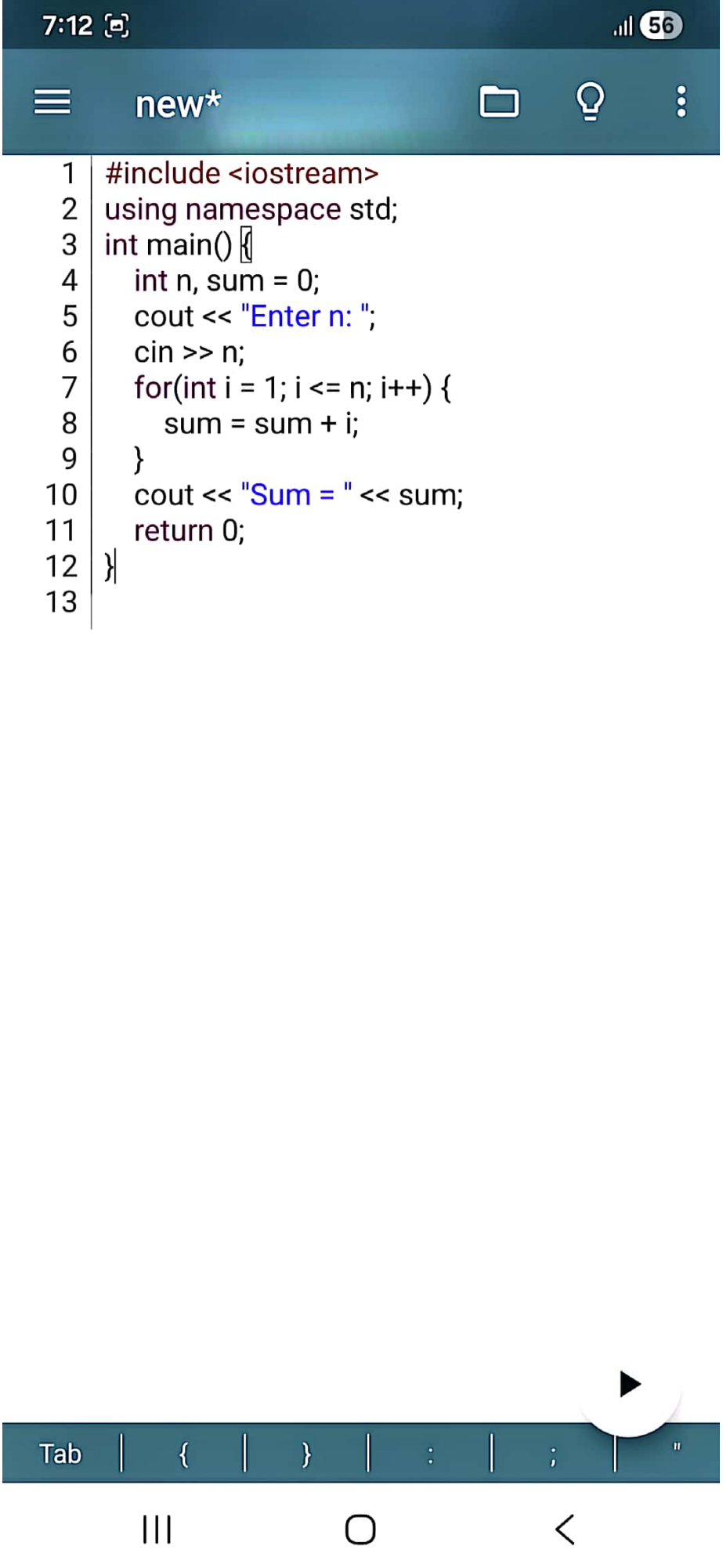
Task:3
sum of natural numbers using loop

```
#include<iostream>
using namespace std;
int main(){
    int n; sum=0;
    cout<<"Enter n=";
    cin>>n;
    for(int i=1;i<=n;i++){
        sum=sum+i;
    }
    cout<<"sum="<<sum;
    return 0;
}
```

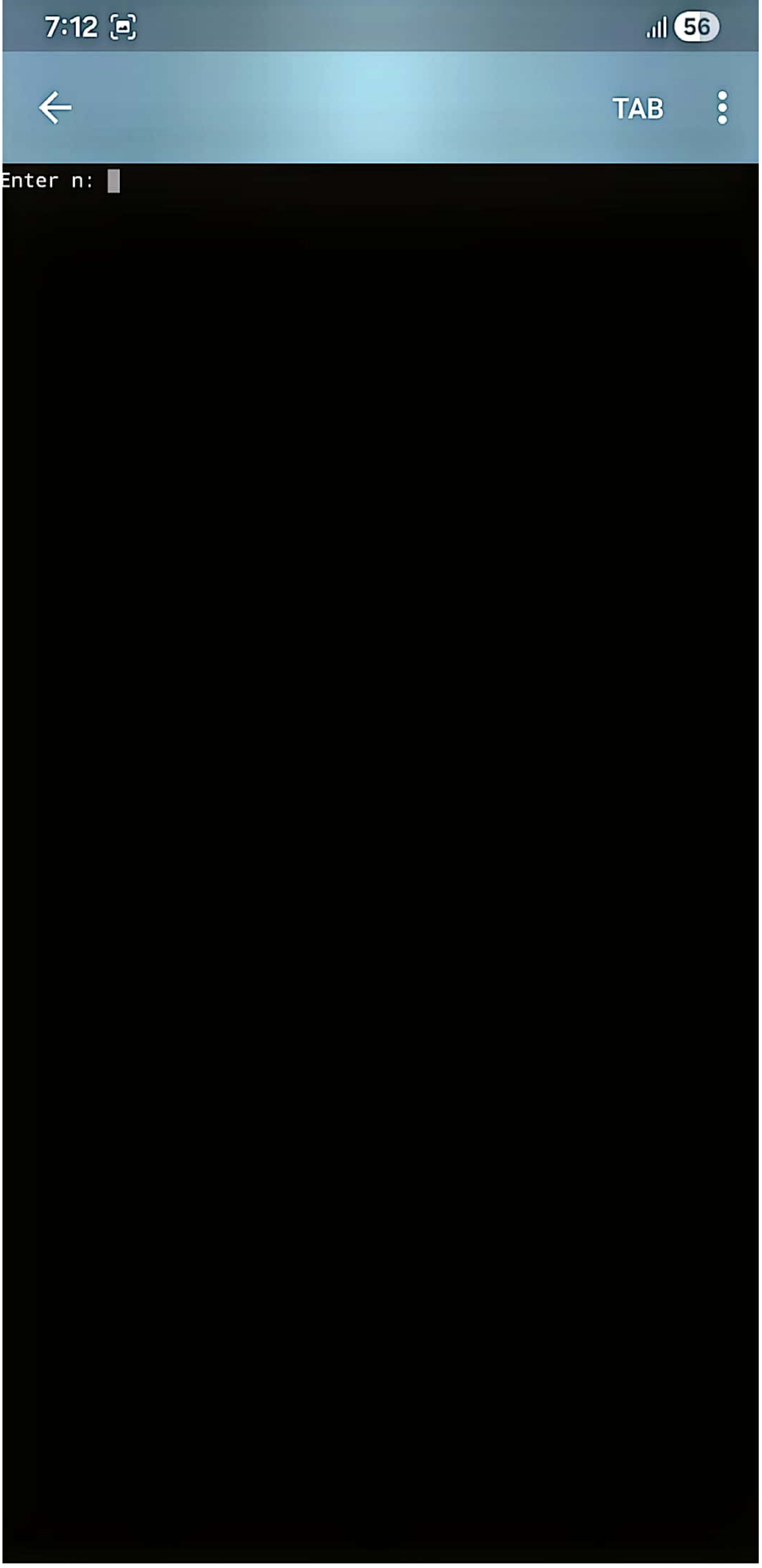
output:-

Enter n-

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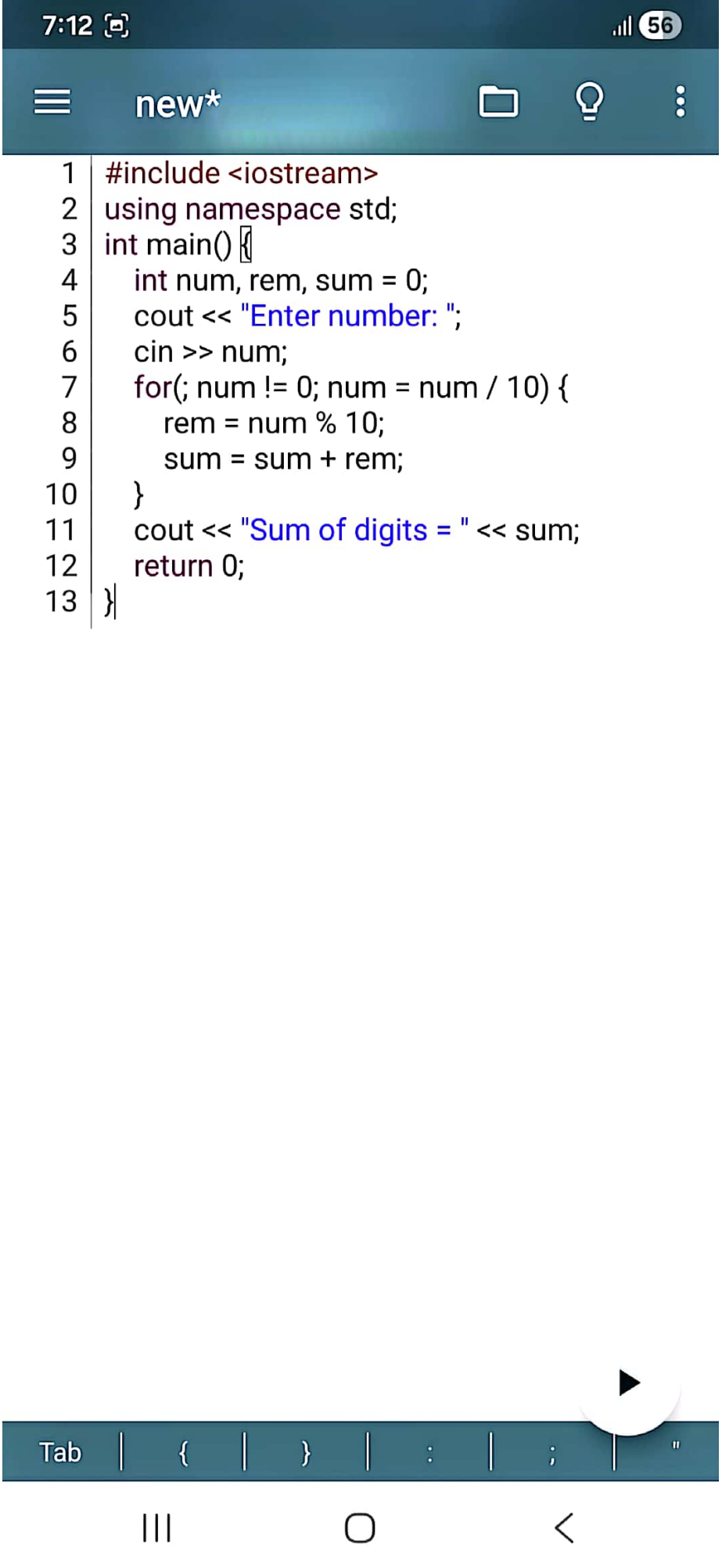
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Task=4
sum of digits of a number

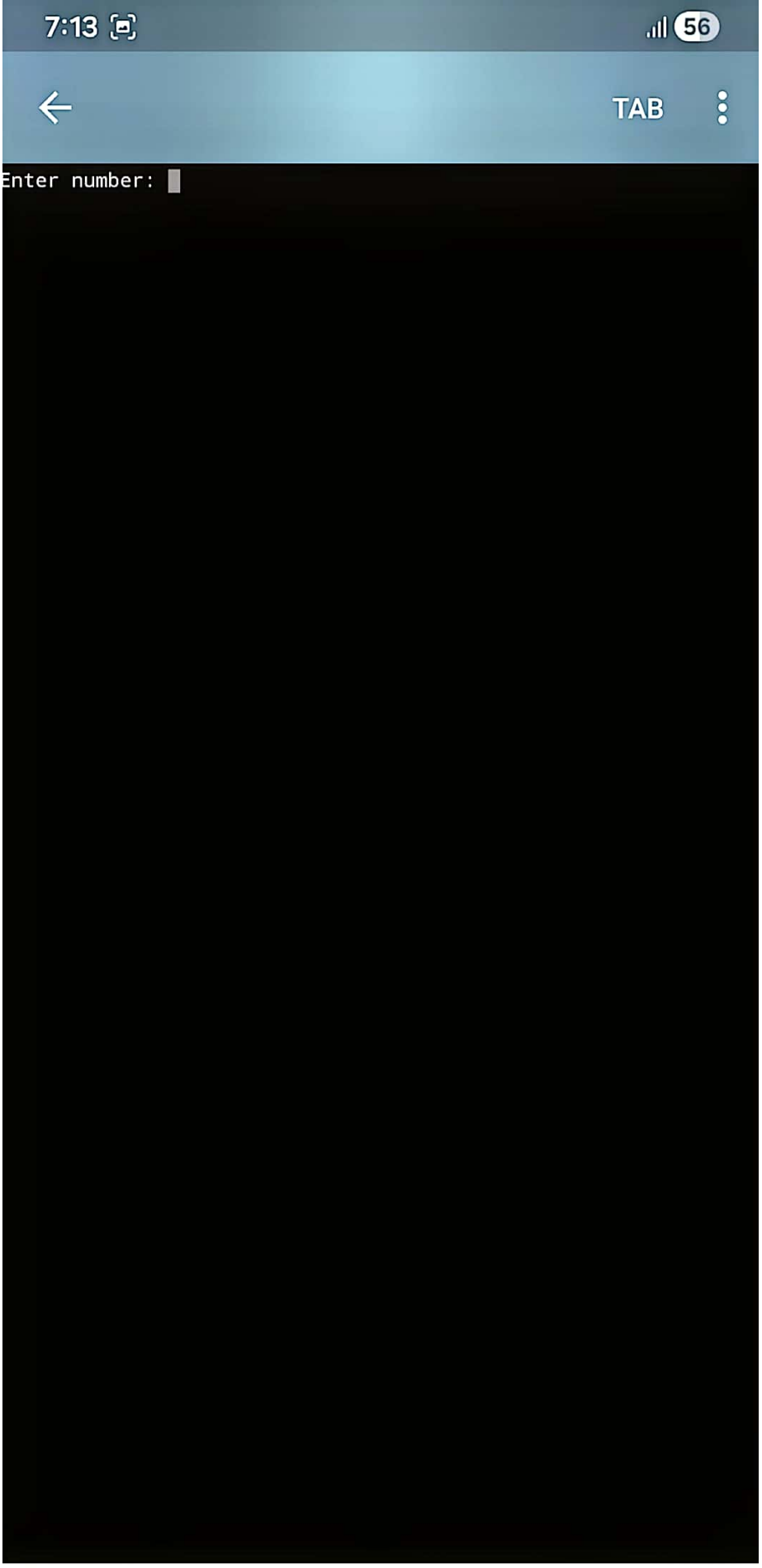
```
#include <iostream>
using namespace std;
int main(){
    int num, rem, sum=0;
    cout<<"Enter number:";
    cin>>num;
    for(int num; num!=0; num=num/10){
        rem= num%10;
        sum = sum+rem;
    }
    cout<<"sum of digits="<<sum;
    return 0;
}
```

output:-

Enter number:



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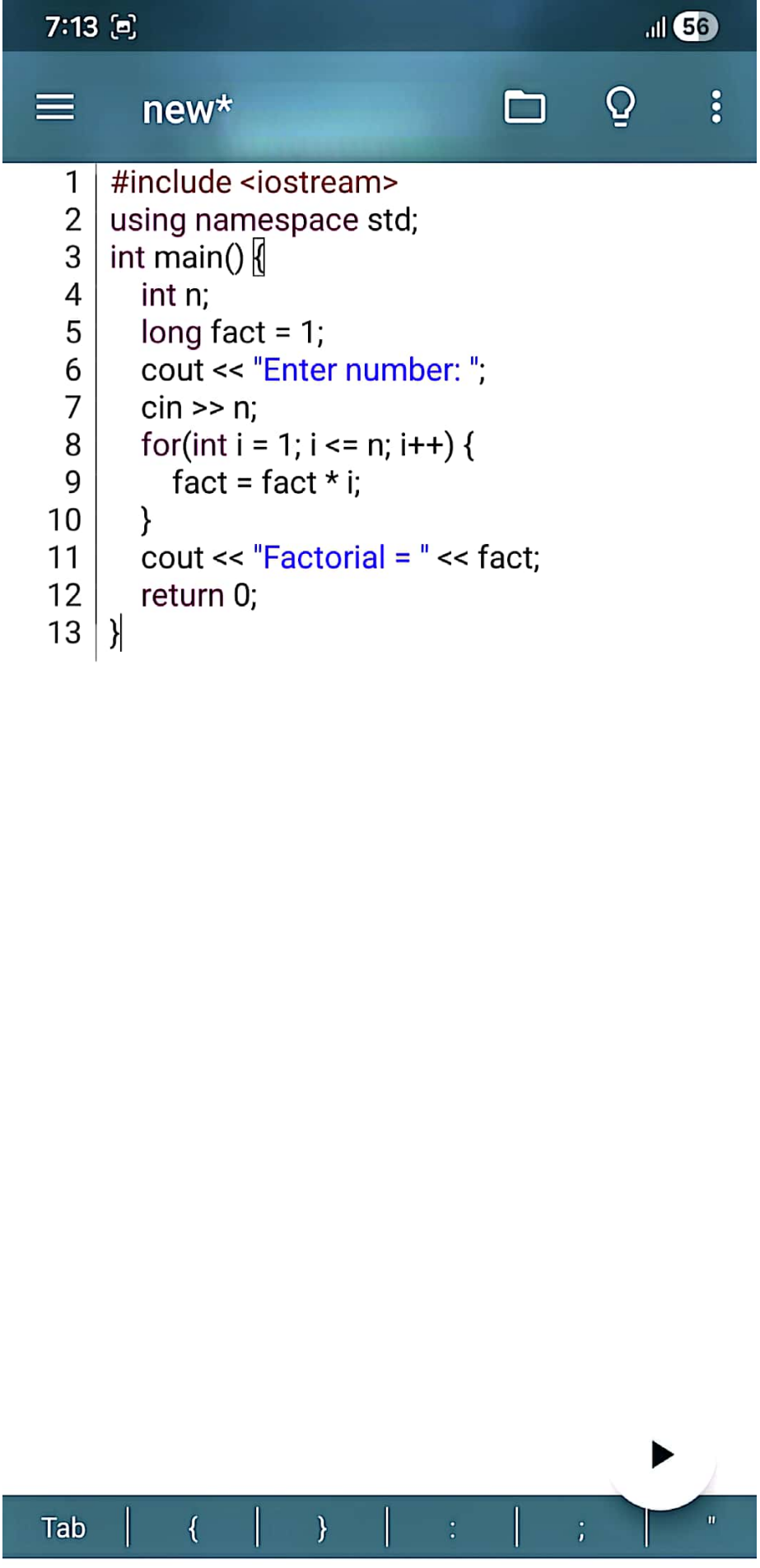
Task 5: factorial of a number

```
#include<iostream>
using namespace std;
int main(){
    int n;
    long fact=1;
    cout<<"Enter a number:";
    cin>>n;
    for(int i=1;i<=n;i++){
        fact=fact*i;
    }
    cout<<"factorial="<<fact;
    return 0;
}
```

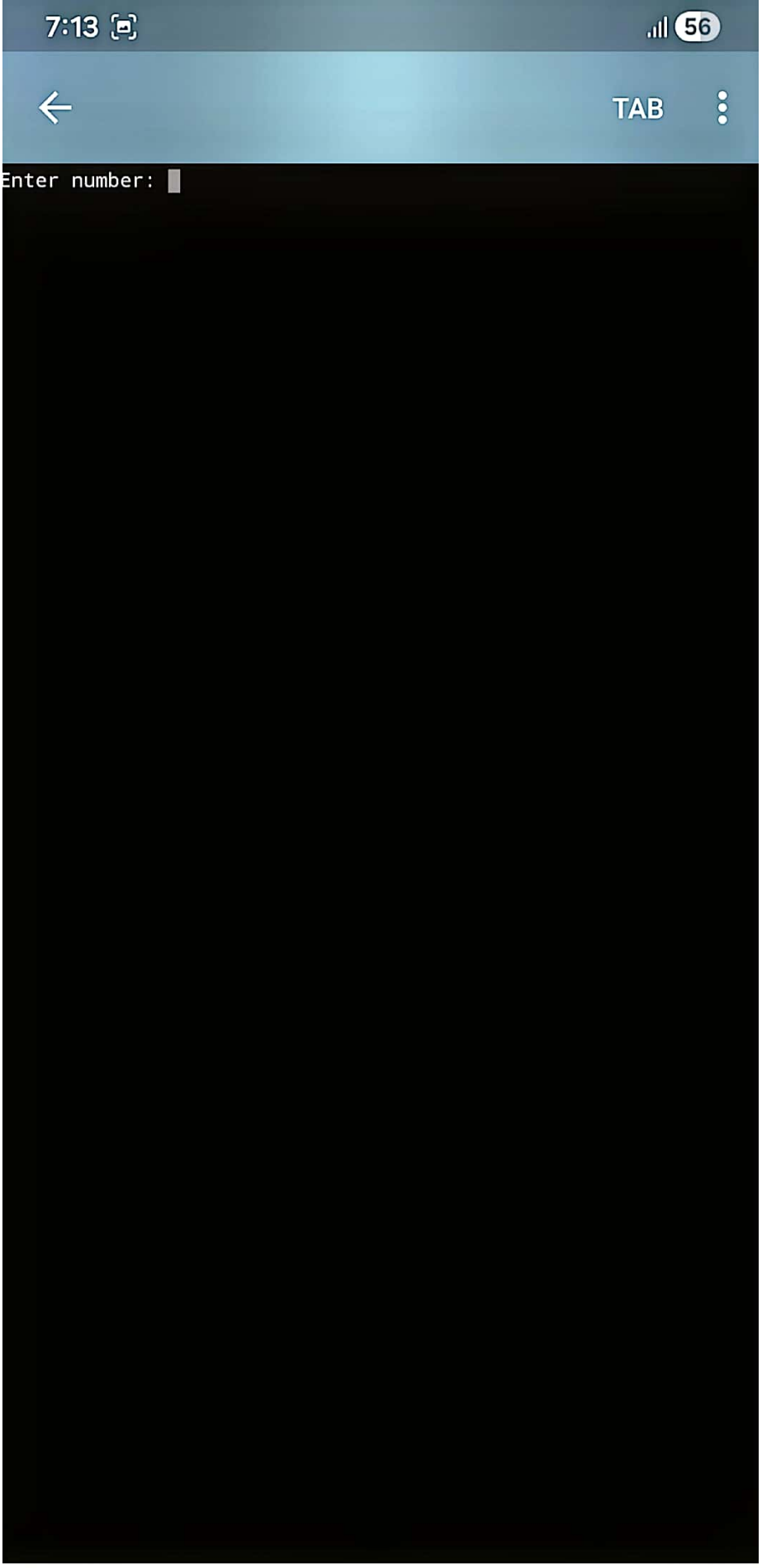
output:-

Enter number:

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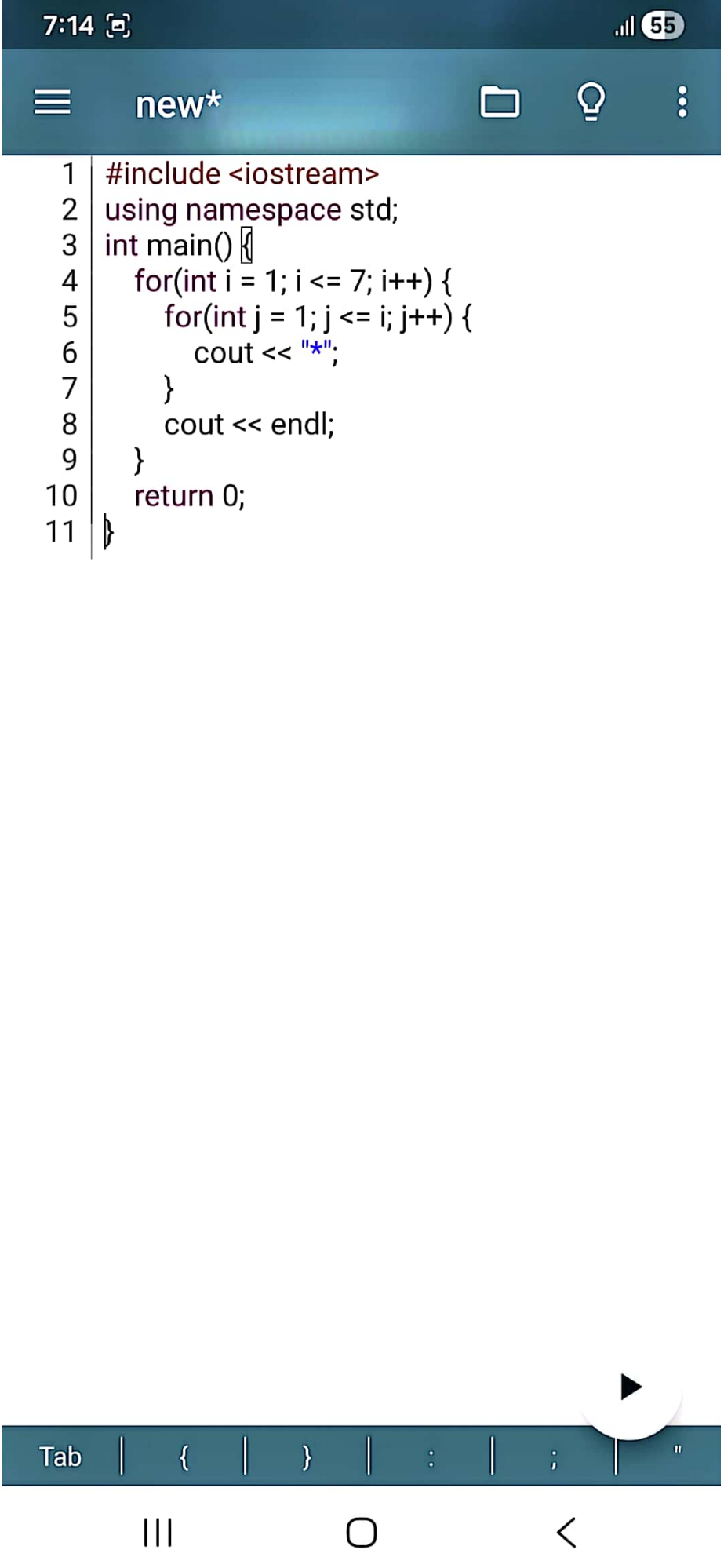
Task: Pattern

```
#include <iostream>
using namespace std;
int main() {
    for(int i=1; i<=7; i++){
        for(int j=1; j<=i; j++){
            cout<<"*";
        }
        cout<<endl;
    }
    return 0;
}
```

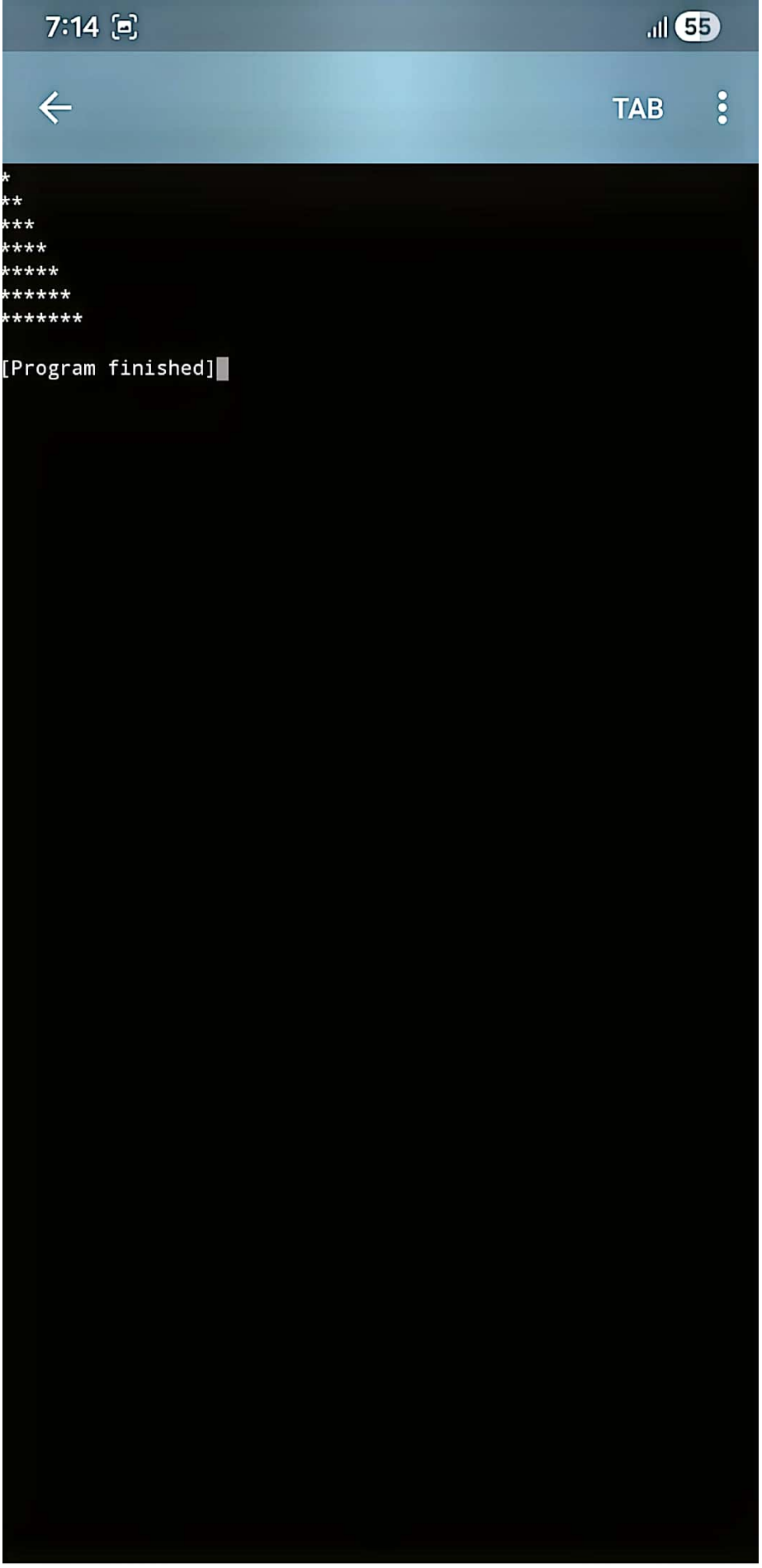
output:-

```
*
**
***
****
*** **
******
****** **
```

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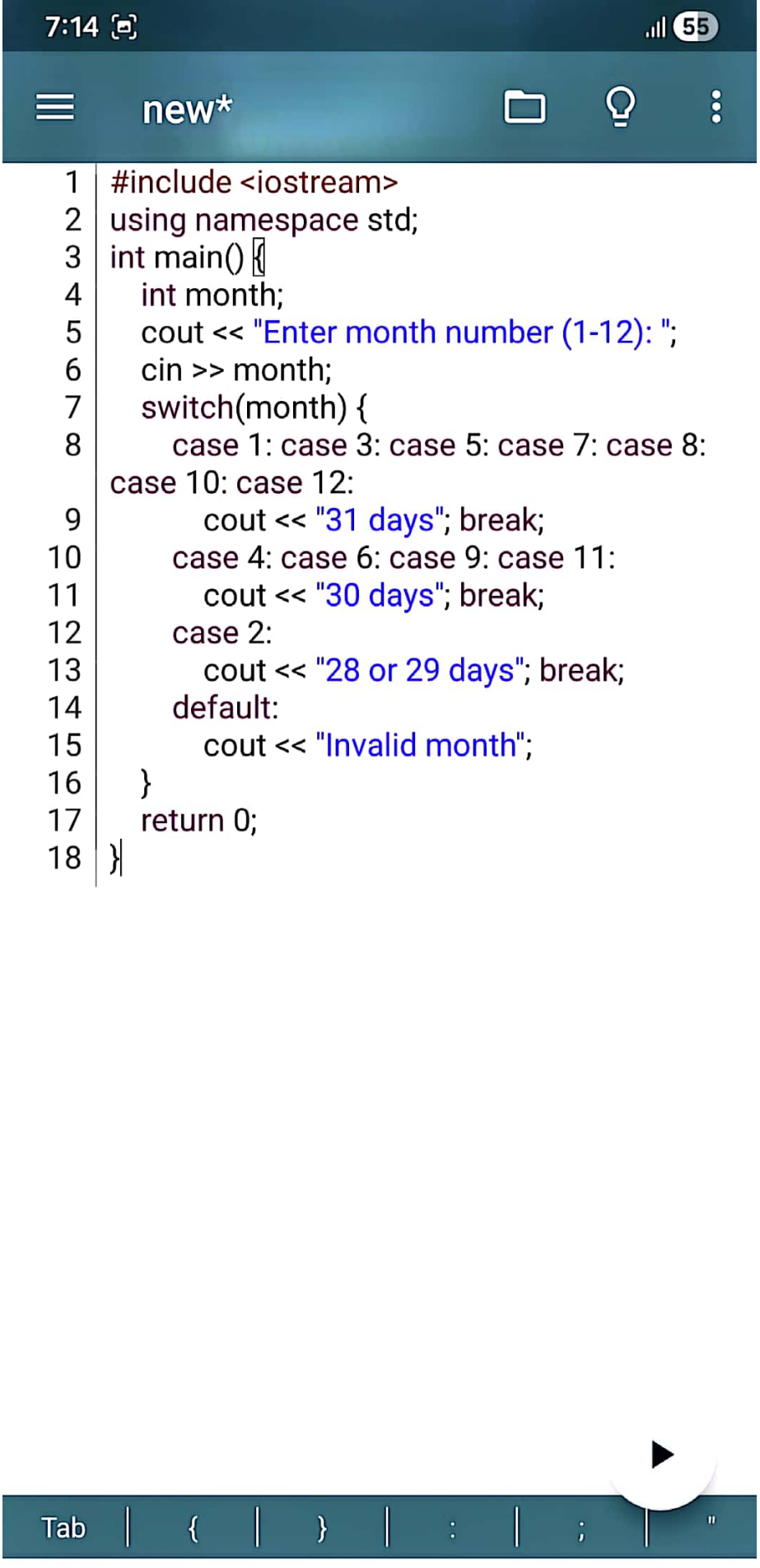


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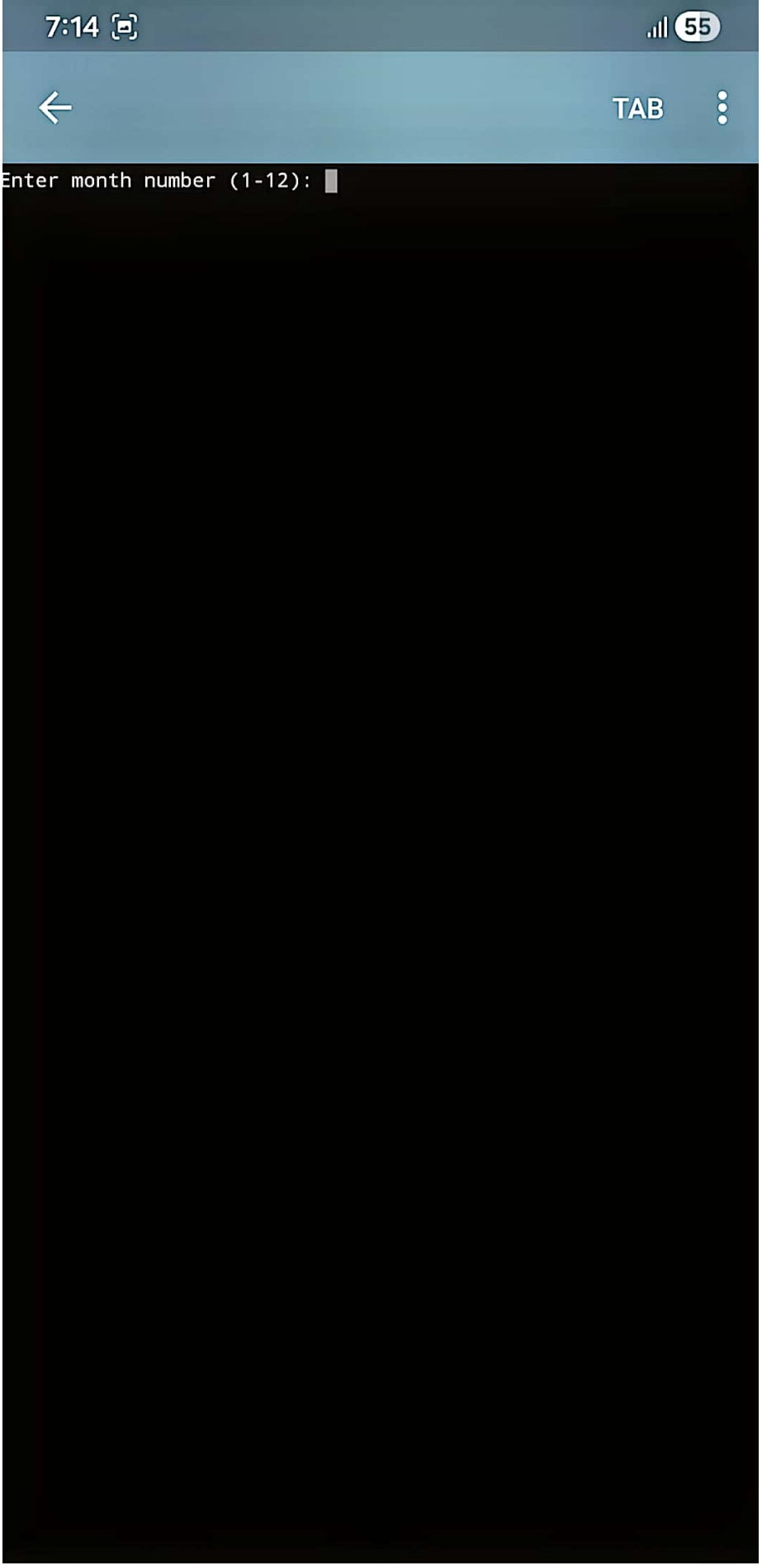
Task = Days in a month using switch case

```
#include<iostream>
using namespace std;
int main(){
    int month;
    cout<<"Enter month number (1-12): ";
    cin>> month;
    switch(month){
        case 1: case 3: case 5: case 7: case 8:
            case 10: case 12:
            cout<< "31 days";
            break;
        case 4: case 6: case 9: case 11:
            cout<<"30 days";
            break;
        case 2:
            cout<<"28 or 29 days";
            break;
        default:
            cout<<"Invalid month";
    }
    return 0;
}
```

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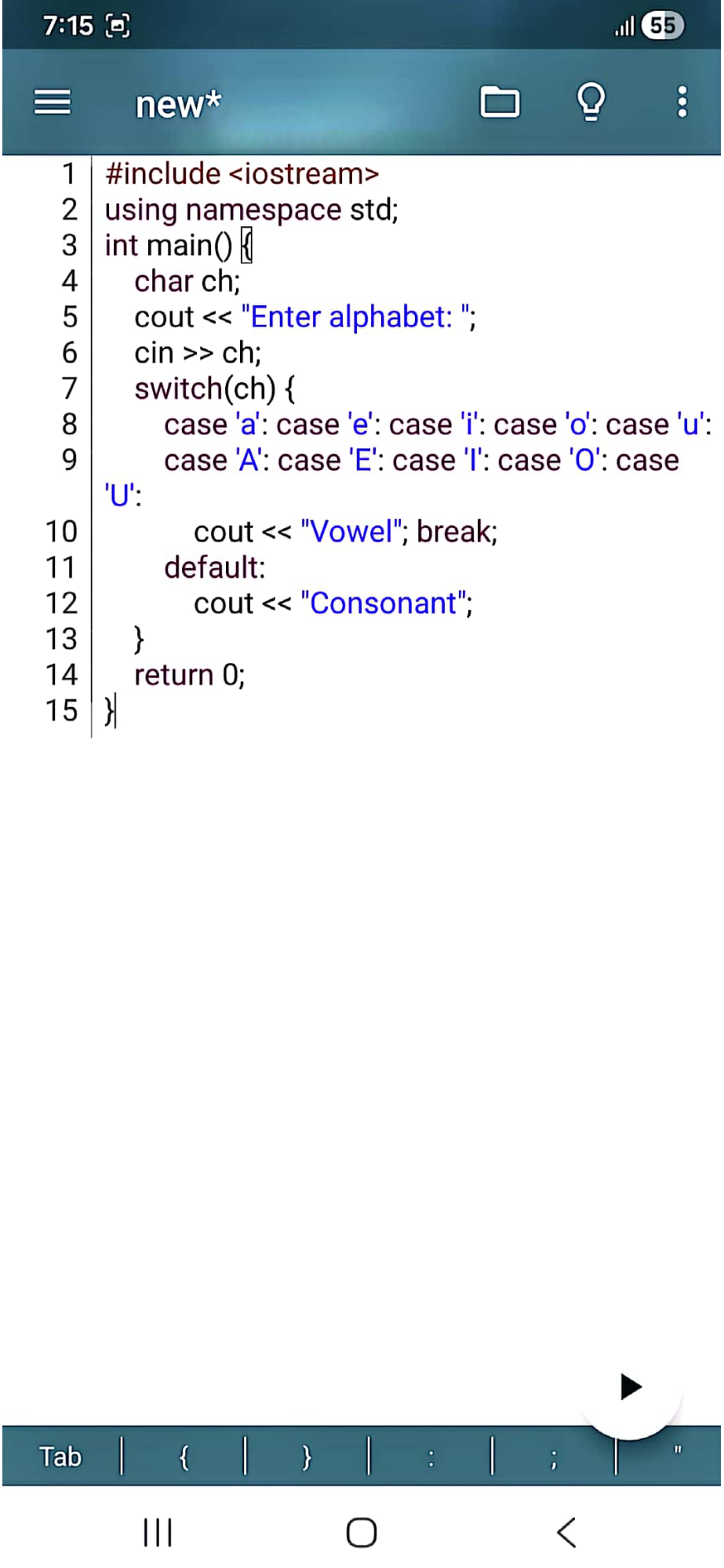
Task: Vowel or Consonant using switch case

```
#include <iostream>
using namespace std;
int main(){
    char ch;
    cout<<"Enter alphabet:";
    cin>>ch;
    switch(ch){
        case 'a': case 'e': case 'i': case 'o': case 'u':
        case 'A': case 'E': case 'I': case 'O': case 'U':
            cout<<"Vowel"; break;
        default:
            cout<<"consonant";
    }
    return 0;
}
```

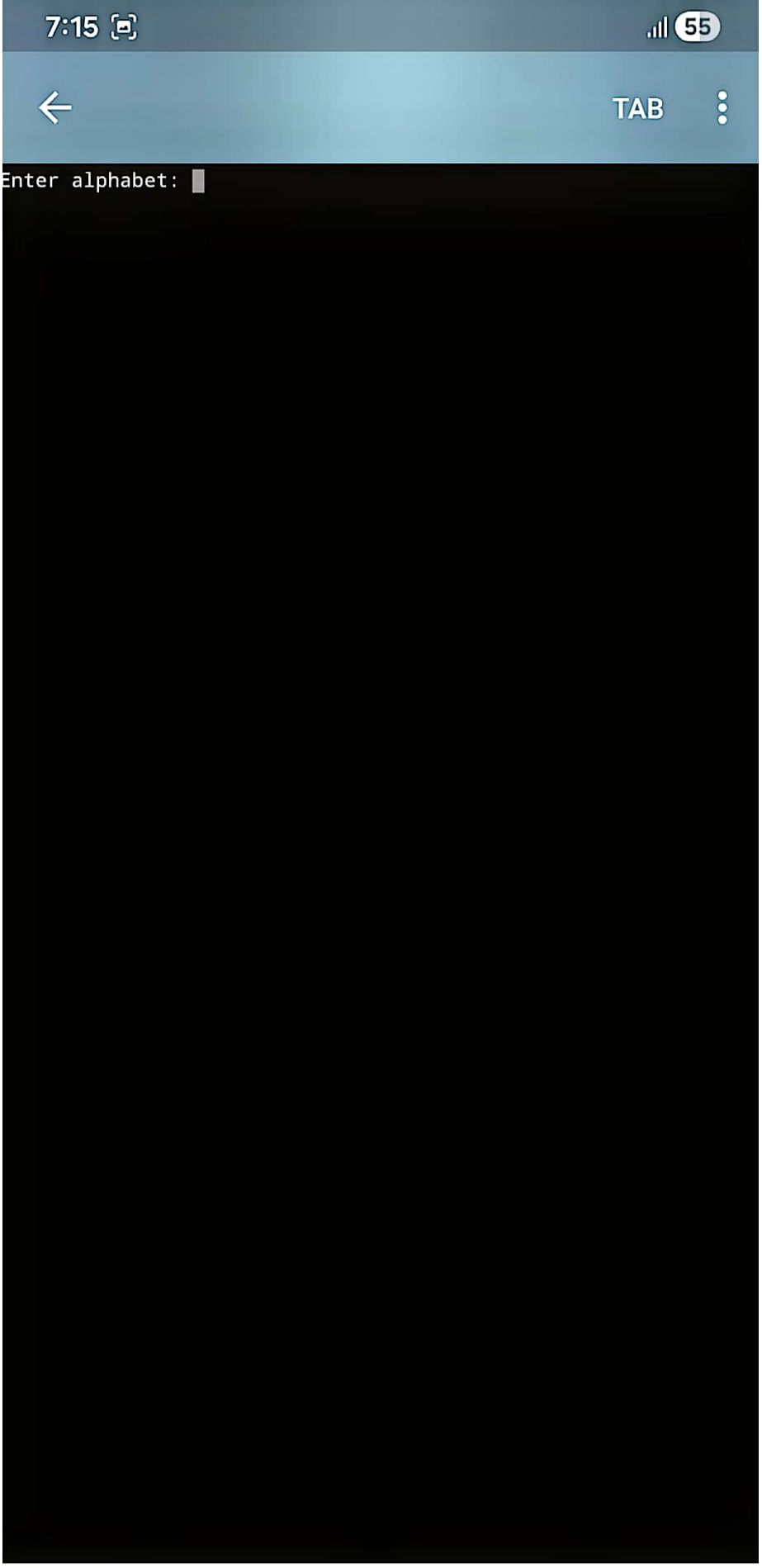
output:-

Enter alphabet-

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